

中山大學

SUN YAT-SEN UNIVERSITY

实验 0x06 实验报告

硬盘驱动与文件系统

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1. MBR 分区表

首先我们需要根据 MBR 的结构定义，利用 `MbrPartition` 的 `parse` 方法来对四个 `partition` 依次解析，如下

```
for i in 0..4 {
  let offset = 0x1BE + i * 16;
  partitions.push(
      // FIXED: parse the mbr partition from the buffer
      //      - just ignore other fields for mbr
      MbrPartition::parse(buffer[offset..offset + 16].try_into().unwrap()),
  );
```

然后我们需要利用 `define_filed!` 完成内部数据的解析提取，由于 `CHS` 的 `start` 和 `end` 中的 `sector` 和 `cylinder` 不是占据整个字节的，因此要另外写 `start_sector` 等函数进行解析提取，部分代码如下

```
define_field!(u8, 0x00, status); // Status (bit 7 set = active or bootable)
define_field!([u8; 3], 0x01, chs_start); // CHS address of first partition sector
(3 bytes)
define_field!(u8, 0x04, partition_type); // Partition type
define_field!([u8; 3], 0x05, chs_end); // CHS address of last partition sector (3
bytes)
define_field!(u32, 0x08, begin_lba); // LBA of partition start
define_field!(u32, 0x0C, total_lba); // Number of sectors in partition

pub fn is_active(&self) -> bool {
    self.status() == 0x80
}

pub fn begin_head(&self) -> u8 {
    self.chs_start()[0]
}

pub fn begin_sector(&self) -> u8 {
    self.chs_start()[1] & 0x3F
}

pub fn begin_cylinder(&self) -> u16 {
    ((self.chs_start()[1] as u16 & 0xC0) << 2) | self.chs_start()[2] as u16
}

// ..... end_head(), end_sector()... is the same as above,
// see entry.rs for the full code
```

然后我们对这部分进行功能测试，需要在根目录的 `Cargo.toml` 的 `member` 中加入 `"crates/storage",` 表项，然后再 `storage/` 下运行 `cargo test`，可以看到，可以通过对应的测试，说明实现大致无误。

```
(base) linkzelda@LAPTOP-QFAEHS2I:~/OS-lab/crates/storage$ cargo test -- --nocapture
running 1 test
Partition Meta Data {
    Active: true,
    Begin Head: "0x01",
    Begin Sector: "0x0001",
    Begin Cylinder: "0x0000",
    Partition Type: "0x0b",
    End Head: "0xfe",
    End Sector: "0x003f",
    End Cylinder: "0x02fc",
    Begin LBA: "0x0000003f",
    Total LBA: "0x00bb867e",
}
test partition::mbr::entry::tests::partition_test ... ok

test result: ok. 1 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out; finished in 0.00
s

   Doc-tests ysos_storage

running 1 test
test crates/storage/src/common/macros.rs - common::macros::define_field (line 5) ... ignored

test result: ok. 0 passed; 0 failed; 1 ignored; 0 measured; 0 filtered out; finished in 0.00
s

all doctests ran in 0.40s; merged doctests compilation took 0.40s
(base) linkzelda@LAPTOP-QFAEHS2I:~/OS-lab/crates/storage$
```